Percentiles, Percentile Ranks, & Interpolation

Pengantar Statistika – Descriptive Statistics Pt 2

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Raw Scores

- Although the primary purpose of a frequency distribution is to provide a description of an entire set of scores, it also can be used to describe the position of an individual within the set.
- Individual scores, or ***X*** **values,** are called raw scores.

Percentile Rank

- The **rank** or **percentile rank** of a particular score is defined as the percentage of individuals in the distribution with scores at or below the particular value.
- When a score is identified by its percentile rank, the score is called a **percentile**.
  - Suppose, for example, that you have a score of $X = 43$ on an exam and that you know that exactly 60% of the class had scores of 43 or lower. Then your score $X = 43$ has a percentile rank of 60%, and your score would be called the 60th percentile.
- Notice that ***percentile rank*** **refers to a percentage** and that ***percentile*** **refers to a score.**

Cumulative Frequency and Cumulative Percentage

- To determine percentiles or percentile ranks, the first step is to find the number of individuals who are located at or below each point in the distribution.
  - This can be done most easily with a frequency distribution table by simply counting the number of scores that are in or below each category on the scale. The resulting values are called ***cumulative frequencies.***

In the following frequency distribution table, we have included a cumulative frequency column headed by *cf*. For each row, the cumulative frequency value is obtained by adding up the frequencies in and below that category. For example, the score $X = 3$ has a cumulative frequency of 14 because exactly 14 individuals had scores of $X = 3$ or less.

X	*F*	*cf*
5	1	20
4	5	19
3	8	14
2	4	6
1	2	2

The cumulative frequencies show the number of individuals located at or below each score. To find percentiles, we must convert these frequencies into percentages. The resulting values are called *cumulative percentages* because they show the percentage of individuals who are accumulated as you move up the scale.

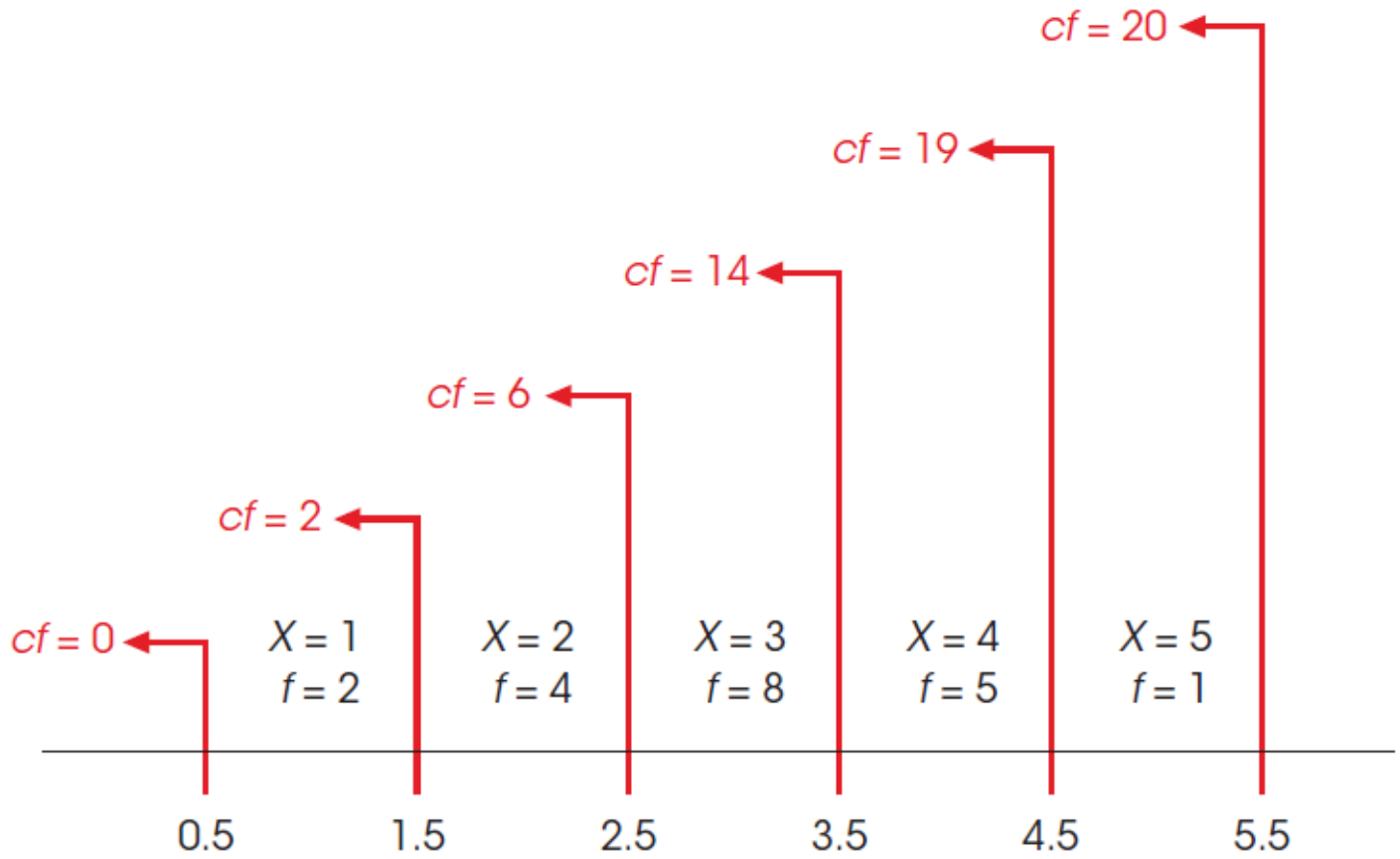
- This time we have added a cumulative percentage column (c%) to the frequency distribution table. The values in this column represent the percentage of individuals who are located in and below each category.
  - For example, 70% of the individuals (14 out of 20) had scores of $X = 3$ or lower.

$$c\% = \frac{cf}{N}(100\%)$$

X	f	Cf	c%
5	1	20	100%
4	5	19	95%
3	8	14	70%
2	4	6	30%
1	2	2	10%

- However, you must remember that the *X* values in the table are usually measurements of a **continuous variable** and, therefore, represent intervals on the scale of measurement.
- A score of ***X* = 2,** for example, means that the measurement was somewhere between the real limits of **1.5 and 2.5.**
- Thus, when a table shows that a score of *X* = 2 has a cumulative percentage of 30%, you should interpret this as meaning that 30% of the individuals have been accumulated by the time you reach the top of the interval for *X* = 2.

FIGURE 2.12
The relationship between cumulative frequencies (*cf* values) and upper real limits. Notice that two people have scores of $X = 1$. These two individuals are located between the real limits of 0.5 and 1.5. Although their exact locations are not known, you can be certain that both had scores below the upper limit of 1.5.



Interpolation

- In many cases, there are many values that do not appear directly in the table, and it is impossible to determine these values precisely.
- For example:
  - What is the 50th percentile?
  - What is the percentile rank for $X = 4$?
- Because these values are not specifically reported in the table, you cannot answer the questions. However, it is possible to estimate these intermediate values by using a procedure known as ***interpolation*.**

$$c\% = \frac{cf}{N}(100\%)$$

X	f	Cf	c%
5	1	20	100%
4	5	19	95%
3	8	14	70%
2	4	6	30%
1	2	2	10%

The general process of interpolation can be summarized as follows:

1. A single interval is measured on two separate scales. The endpoints of the interval are known for each scale.
2. You are given an intermediate value on one of the scales. The problem is to find the corresponding intermediate value on the other scale.

3. The interpolation process requires four steps:
    1. Find the width of the interval on both scales.
    2. Locate the position of the intermediate value in the interval. This position corresponds to a fraction of the whole interval:

$$\text{fraction} = \frac{\text{distance from the top of the interval}}{\text{interval width}}$$

    3. Use the same fraction to determine the corresponding position on the other scale. First, use the fraction to determine the distance from the top of the interval: $\text{distance} = (\text{fraction}) \times (\text{width})$
    4. Use the distance from the top to determine the position on the other scale.

Example 1

Using the following distribution of scores, we will find the percentile rank corresponding to $X = 7.0$:

Hint: Continuous, use upper real limit

X	f	cf	c%
10	2	25	100%
9	8	23	92%
8	4	15	60%
7	6	11	44%
6	4	5	20%
5	1	1	4%

Notice that $X = 7.0$ is located in the interval bounded by the real limits of 6.5 and 7.5. The cumulative percentages corresponding to these real limits are 20% and 44%, respectively. These values are shown in the following table:

	Scores (X)	Percentages
Top	7.5	44%
Intermediate value →	7.0	?
Bottom	6.5	20%

For interpolation problems, it is always helpful to create a table showing the range on both scales.

STEP 1 For the scores, the width of the interval is 1 point (from 6.5–7.5). For the percentages, the width is 24 points (from 20–44%).

STEP 2 Our particular score is located 0.5 point from the top of the interval. This is exactly halfway down in the interval.

STEP 3 On the percentage scale, halfway down is

$$\frac{1}{2}(24 \text{ points}) = 12 \text{ points}$$

STEP 4 For the percentages, the top of the interval is 44%, so 12 points down would be

$$44\% - 12\% = 32\%$$

This is the answer. A score of $X = 7.0$ corresponds to a percentile rank of 32% ■

Example 2

Using the following distribution of scores, we will use interpolation to find the 40th percentile:

X	*F*	*cf*	*c%*
20–24	2	20	100%
15–19	3	18	90%
10–14	3	15	75%
5–9	10	12	60%
0–4	2	2	10%

A percentage value of 40% is not given in the table; however, it is located between 10% and 60%, which are given. These two percentage values are associated with the upper real limits of 4.5 and 9.5, respectively. These values are shown in the following table:

	Scores (X)	Percentages	
Top	9.5	60%	
	?	40%	← Intermediate value
Bottom	4.5	10%	

STEP 1 For the scores, the width of the interval is 5 points. For the percentages, the width is 50 points.

STEP 2 The value of 40% is located 20 points from the top of the percentage interval. As a fraction of the whole interval, this is 20 out of 50, or $\frac{2}{5}$ of the total interval.

STEP 3 Using this same fraction for the scores, we obtain a distance of

$$\frac{2}{5}(5 \text{ points}) = 2 \text{ points}$$

The location we want is 2 points down from the top of the score interval.

STEP 4 Because the top of the interval is 9.5, the position we want is

$$9.5 - 2 = 7.5$$

This is the answer. The 40th percentile is $X = 7.5$. ■

Exercise

Using the following distribution of scores, we will use interpolation to find the 40th percentile:

X	F	cf	c%
20–24	2	20	100%
15–19	3	18	90%
10–14	3	15	75%
5–9	10	12	60%
0–4	2	2	10%

1. In a distribution of exam scores, which of the following would be the highest score?

- **a.** the 20th percentile
- **b.** the 80th percentile
- **c.** a score with a percentile rank of 15%
- **d.** a score with a percentile rank of 75%

2. Following are three rows from a frequency distribution table. For this distribution, what is the 90th percentile?

- **a.** X = 24.5
- **b.** X = 25
- **c.** X = 29
- **d.** X = 29.5

X	c%
30–34	100%
25–29	90%
20–24	60%

3. Following are three rows from a frequency distribution table. Using interpolation, what is the percentile rank for $X = 18$?

- **a.** 52.5%
- **b.** 30%
- **c.** 29%
- **d.** 25%

X	c%
20–24	60%
15–19	35%
10–14	15%

Thank You